

- 1 (a) A ball leaves the edge of a table with a horizontal velocity v , as shown in Fig. 1.1.

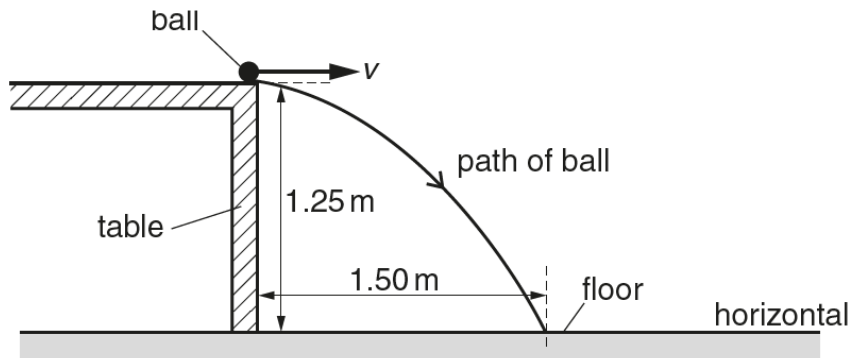


Fig. 1.1

The height of the table is 1.25 m. The ball travels a distance of 1.50 m horizontally before hitting the floor.

Air resistance is negligible.

Calculate, for the ball,

- (i) the horizontal velocity v as it leaves the table,

$$u_x = v, u_y = 0$$

$$s_x = u_x t = vt \quad [1]$$

$$s_y = 0 + \frac{1}{2}at^2 \Rightarrow 1.25 = 0 + \frac{1}{2}(9.81)t^2 \quad [1]$$

$$v = \dots\dots\dots \text{ m s}^{-1} \quad [2]$$

- (ii) the velocity just before it hits the floor.

$$v = \sqrt{(v_y)^2 + (v_x)^2} \quad [1]$$

$$v_y = u_y + at = 0 + (9.81)(0.505) = 4.95 \text{ m s}^{-1} [1]$$

$$\text{velocity} = \dots\dots\dots \text{ m s}^{-1} \quad [2]$$

(b) A second ball leaves the edge of the table with a horizontal velocity $2v$.

- (i) State and explain whether the time taken to hit the floor is the same or different compared to the first ball.

Since both ball initial velocity for the vertical component is zero, the height is the same and same acceleration of free fall, the time taken is the same.

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.....[2]

- (ii) Describe the variation of the vertical component of velocity if air resistance is not negligible.

The vertical component of velocity is increasing at a decreasing rate.

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.....[2]

[Total: 8]

2 (a) State Newton's Second Law of motion